ENGR 1451/2451: Exploratory Data Science

The Chi-Square Test

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Today: The Chi-Square Test

A chi-square test asks whether the frequencies we observed differ more from expectations than chance variation can explain.

- Test whether observations follow a specified category distribution (**goodness of fit**).
- Compute expected frequencies and the χ^2 statistic.
- Interpret degrees of freedom and a right-tail P-value.
- Test whether two categorical variables are **independent**.

Is This Die Fair?

A gambler rolls a die 60 times

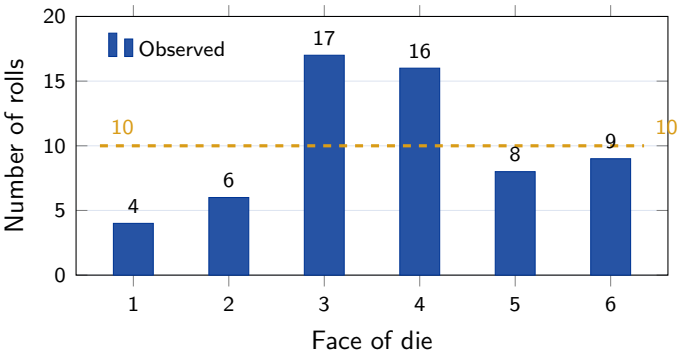
Face	Number of times
1	4
2	6
3	17
4	16
5	8
6	9
Total	60

The claim

The gambler says the die is fair: all six faces have the same chance.

Would these results surprise you if the die were fair?

Observed Counts vs. Expected Counts



If the die is fair

$$P(1) = \cdots = P(6) = \frac{1}{6}$$

$$E_i = 60\left(\frac{1}{6}\right) = 10$$

Expected: ten of each face.

Need a **single statistic** that combines the differences between *all* observed and expected frequencies.

The χ^2 Statistic

Combine discrepancies across all categories

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

O_i = observed frequency, E_i = expected frequency

Why square the differences?

Positive and negative differences do not cancel. Larger deviations count more.

Why divide by expected count?

A given discrepancy matters more in a category where relatively few observations are expected.

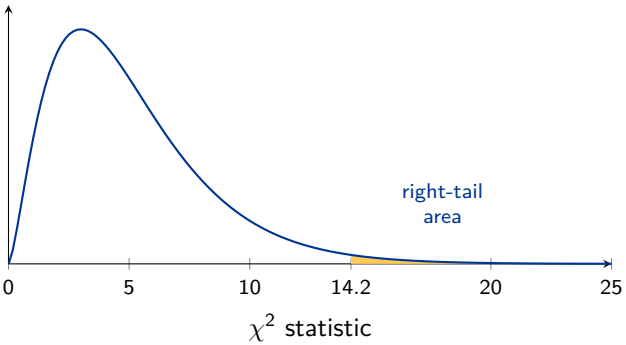
Large χ^2 means observed frequencies are far from expectations.

Worked Example: 60 Rolls of the Die

Face	Observed O	Expected E	$O - E$	$(O-E)^2/E$
1	4	10	-6	3.6
2	6	10	-4	1.6
3	17	10	$+7$	4.9
4	16	10	$+6$	3.6
5	8	10	-2	0.4
6	9	10	-1	0.1
Total	60	60	0	**14.2**

How Unusual Is $\chi^2 = 14.2$?

χ^2 curve, 5 degrees of freedom (since six categories)



For a fair die

- $P(\chi^2_5 \geq 14.2)$

$$P \approx 0.0144$$

Why the right tail? Larger χ^2 values mean even worse agreement with the fair-die model.

Where Do the Degrees of Freedom Come From?

If category probabilities are fully specified

$$\text{degrees of freedom} = k - 1$$

Here, $k = 6$ faces $\Rightarrow \mathrm{df} = 6 - 1 = \boxed{5}$.

4	6	17	16	8	9

Five counts can vary; the sixth is fixed by the total of 60.

What Does the P-value Mean?

The correct interpretation

If the die were fair, the chance of obtaining a χ^2 statistic of 14.2 or larger in 60 rolls is about **1.4%**.

Null hypothesis H_0

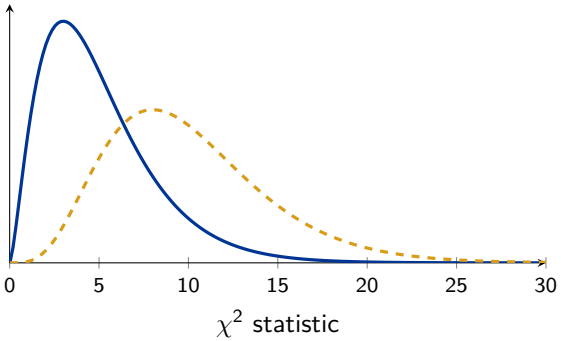
The six faces are equally likely.

Alternative H_A

At least one face has a different probability.

At the 5% significance level, **reject** H_0. There is evidence that the die is not fair.

Discussion: Reading Chi-Square Curves



Without calculating:

1. Which curve is $\mathrm{df} = 5$? Which is $\mathrm{df} = 10$?
2. At $\chi^2 = 15$, which has the larger *P*-value: $\mathrm{df} = 5$ or $\mathrm{df} = 10$?
3. With $\mathrm{df} = 10$, which gives a larger *P*-value: $\chi^2 = 15$ or 20?

When Should You Use a χ^2 Test?

χ^2 goodness-of-fit test

Are the category frequencies consistent with specified probabilities?

z- or t-test

Is a population *mean* consistent with a specified value?

Conditions for the ordinary goodness-of-fit approximation

- Data are counts from random trials or an appropriate random sample.
- As a rule of thumb, each expected count is at least 5.

California's Lottery Wheel: Beyond the P-Value

The statistical test

A wheel with **100 slots** was spun **800 times**.

$$\chi^2 = 119, \quad \mathrm{df} = 99, \quad P \approx 8\%$$

The physical investigation

1. Slot 69 appeared most often; opposite slot 19 appeared least often.
2. A balancing weight was found near slot 69 and removed.
3. The wheel was rebalanced and retested.
4. During retesting, the wheel squeaked, so it was oiled. Later results looked better.

Statistical insignificance does not prove fairness. Physical evidence can still justify investigation.

Are Coating Method and Inspection Outcome Related?

An engineering team inspects **240 synthetic coating coupons** made by Dip or Spray coating. Each coupon is classified as Pass, Pinholes, or Delamination.

Observed inspection counts

Method	Pass	Pinholes	Delamination	Total
Dip	65	20	15	100
Spray	112	15	13	140
Total	177	35	28	240

The question

Are **coating method** and **inspection outcome** associated, or could the sample differences be due to chance?

First Compare Percentages Within Each Method

Method	Pass	Pinholes	Delamination	Total
Dip ($n = 100$)	65.0%	20.0%	15.0%	100%
Spray ($n = 140$)	**80.0%**	10.7%	9.3%	100%

What we observe

The Spray group has a **higher pass rate**: 80% vs. 65%.

What the test asks

Could discrepancies this large arise by chance *if method and outcome are independent*?

Testing Independence: Expected Counts

Null hypothesis H_0

Coating method and inspection outcome are **independent**.

Alternative H_A

The inspection-outcome distribution **differs** between the two methods.

If H_0 is true

The expected count in each cell is

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{N}.$$

Example: Dip-coated coupons passing inspection

$$E = \frac{(100 \text{ Dip})(177 \text{ Pass})}{240 \text{ coupons}} = \boxed{73.75}.$$

Coating Inspection: Observed vs. Expected

Method	Pass	Pinholes	Delamination
Dip ($n = 100$)	65 (73.75)	20 (14.58)	15 (11.67)
Spray ($n = 140$)	112 (103.25)	15 (20.42)	13 (16.33)

Each cell shows **observed** (**expected**) count under H_0.

Combine discrepancies across all six cells

$$\chi^2 = \sum_{\text{six cells}} \frac{(O - E)^2}{E}.$$

All expected counts exceed 5; use *unrounded* values in calculations.

Question: Which cell contributes most to χ^2?

Test Result: Coating Method and Inspection Outcome

Chi-square statistic

Using the unrounded expected counts:

$$\boxed{\chi^2 \approx 6.86}$$

With $r = 2$ rows (methods) and $c = 3$ columns (outcomes),

$$\mathrm{df} = (r-1)(c-1) = \boxed{2}.$$

Once the row and column totals are known, not every cell count can vary independently.

Right-tail probability

$$P(\chi^2_2 \geq 6.86) \approx \boxed{0.0324}$$

Since $P < 0.05$, **reject independence** at the 5% level.

Evidence of an association between coating method and inspection outcome in this synthetic example.

What Does the Coating Test Tell Us?

- The χ^2 test indicates an **association** between the two categorical variables: method (Dip or Spray) and inspection outcome (Pass, Pinholes, Delamination).
- A significant result does **not establish causation**. Differences in materials, processing conditions, or selection could also matter.
- Valid inference requires independent observations and an appropriate sampling or experimental design. **These data are synthetic** for teaching.
- Compare **percentages within each method**, not only raw counts: group sample sizes differ.
- If groups were sampled separately, the same calculation is called a χ^2 **homogeneity** test.

Degrees of freedom depend on the question

Goodness of fit (fully specified): $\mathrm{df} = k - 1$ Independence: $\mathrm{df} = (r-1)(c-1)$.

Chi-Square Tests in Python

Independence: coating inspections

```
import pandas as pd
from scipy import stats

data = pd.read_csv("coating_inspections.csv")
counts = pd.crosstab(
    data["coating_method"],
    data["inspection_result"])

result = stats.chi2_contingency(counts)
print(result.statistic, result.pvalue)
```

$$\chi^2 = 6.86, \quad \mathrm{df} = 2, \quad P = 0.0324$$

Goodness of fit: fair die

```
observed = [4, 6, 17, 16, 8, 9]
expected = [10, 10, 10, 10, 10, 10]

result = stats.chisquare(
    f_obs=observed,
    f_exp=expected)

print(result.statistic, result.pvalue)
```

$$\chi^2 = 14.2, \quad \mathrm{df} = 5, \quad P = 0.0144$$

`chisquare()` tests a specified distribution; `chi2_contingency()` tests independence and automatically computes expected counts.

Can Observations Be Too Close to Expectations?

Fisher revisited Mendel's experiments

Pooling independent breeding trials gave:

$$\chi^2 < 42, \qquad \mathrm{df} = 84.$$

Their agreement with the predicted ratios was astonishingly close.

A different alternative

If reported frequencies had been adjusted to better match expectations, unusually small χ^2 values would be suspicious.

$$P(\chi^2_{84} < 42) \approx 0.00004$$

Normally: large $\chi^2 \Rightarrow$ right tail. Here: small $\chi^2 \Rightarrow$ left tail.

Summary: Which Statistical Test Should You Use?

What are you testing?	Statistical test
Numerical response: means	
One mean versus a target; population σ known	One-sample z-test
One mean versus a target; σ unknown	One-sample t-test
Two *independent* group means	Two-sample t-test (Welch)
Before/after measurements on the *same* units	Paired t-test
Categorical response: counts	
Observed counts versus specified category probabilities	χ^2 goodness-of-fit
Association between two categorical variables	χ^2 independence
Compare category distributions from different groups	χ^2 homogeneity

Summary

- The χ^2 test compares observed and expected counts across categories:

$$\chi^2 = \sum \frac{(O - E)^2}{E}.$$

- **Goodness of fit** tests a specified category distribution; $\mathrm{df} = k - 1$ for a fully specified model.
- **Independence** tests an association between categorical variables; expected counts use row and column totals, and $\mathrm{df} = (r - 1)(c - 1)$.
- A right-tail P-value asks how often a chance model produces this much discrepancy or more; it does *not* give the probability the null is true.
- Always consider sampling assumptions, expected counts, and what the result cannot establish.